COVID-19 Global Impact: what the country numbers really tell us

Executive summary | Data: Worldometer country snapshot, 226 countries and territories (appears to date from about mid-May 2022; the file itself has no date) | Analyst: Radhika Dinesh Shet

521 M	6.29 M	1.21%	226
reported cases	reported deaths	crude fatality rate	countries & territories

The question

Countries report very different COVID-19 numbers. **How much of that difference is real, and how much is just how hard each country looked for cases?** Everything is normalised by population, and testing intensity is used as the lens.

Five findings

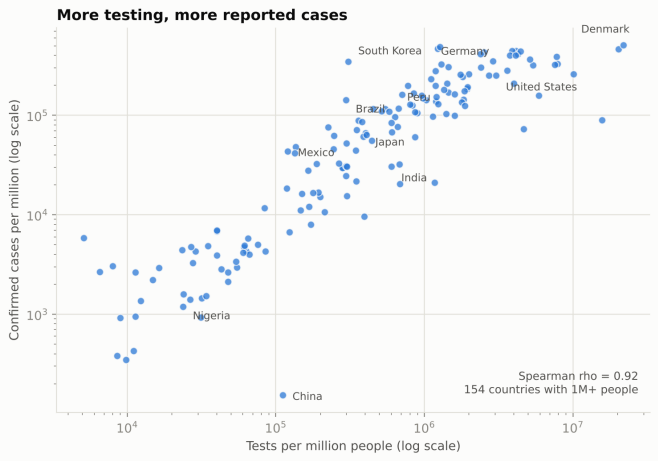
1. The burden is very uneven. Africa has about 18% of the people in the data but about 2% of reported cases and 4% of deaths. Europe has about 9% of the people but 37% of cases and 29% of deaths. South America has the highest deaths per million (about 2,960).

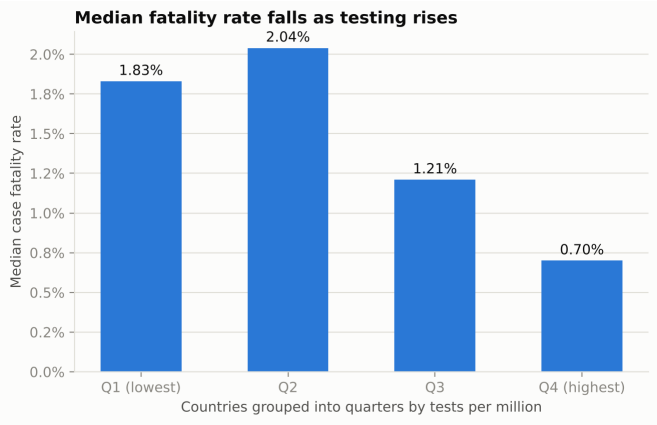
2. Reported cases mostly track testing. Tests per million and cases per million move almost together (correlation 0.92 across 154 countries). The least-testing quarter reports a median of ~3,000 cases per million; the most-testing quarter ~260,000.

3. Fatality rates look worst where testing is thinnest. Median case fatality is 1.8% in the least-testing quarter versus 0.7% in the most-testing quarter, a sign that cases are being missed, not only that deaths are higher.

4. Deaths per million is the fairer yardstick. The heaviest reported toll is in Peru (~6,300 per million), Bulgaria, Bosnia and Herzegovina, Hungary and North Macedonia. The lowest figures are mostly low-testing countries where under-reporting is likely.

5. High positivity with little testing is a warning sign. Brazil (48% of tests positive), Mexico, Ecuador, Bolivia, Honduras, Syria and Eritrea all combine positivity above 30% with fewer than 300,000 tests per million. Their true toll is probably higher than reported.





What to do with this

Compare countries on **deaths per million**; read **cases per million** and **fatality rate** only alongside testing. Treat low numbers in low-testing countries as 'lowest reported', not 'lowest real'.

Limits

One snapshot, so no waves, vaccination effects, rolling averages or doubling times. Reporting rules differ by country. Recovered/active cases were unreliable for 38 countries (blank or implausibly low) and were not used. 69 countries under 1 million people are excluded from rankings. Correlations show association, not cause.